

Capacitance

JEE Main 2020 Chapterwise

Questions with Answer Keys

Physics

Q1 JEE Main 2020 - 2 September (Morning)

A $5\mu\text{F}$ capacitor is charged fully by a 220V supply. It is then disconnected from the supply and is connected in series to another uncharged $2.5\mu\text{F}$ capacitor. If the energy change during the charge redistribution is $\frac{X}{100}\text{ J}$ then value of X to the nearest integer is _____.

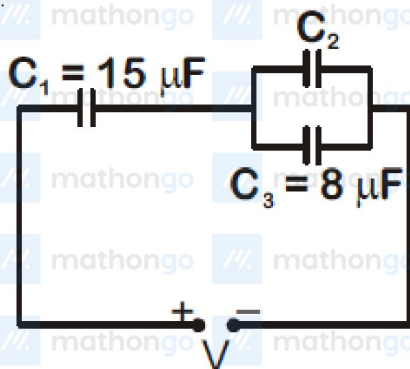
Q2 JEE Main 2020 - 2 September (Evening)

A $10\mu\text{F}$ capacitor is fully charged to a potential difference of 50V . After removing the source voltage it is connected to an uncharged capacitor in parallel. Now the potential difference across them becomes 20V . The capacitance of the second capacitor is

- (A) $20\mu\text{F}$
- (B) $10\mu\text{F}$
- (C) $15\mu\text{F}$
- (D) $30\mu\text{F}$

Q3 JEE Main 2020 - 3 September (Morning)

In the circuit shown in the figure, the total charge is $750\mu\text{C}$ and the voltage across capacitor C_2 is 20V . Then the charge on capacitor C_2 is



- (A) $650\mu\text{C}$
- (B) $590\mu\text{C}$
- (C) $160\mu\text{C}$
- (D) $450\mu\text{C}$

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Q4 JEE Main 2020 - 4 September (Evening)

A capacitor C is fully charged with voltage V_0 . After disconnecting the voltage source, it is connected in parallel with another uncharged capacitor of capacitance $\frac{C}{2}$. The energy loss in the process after the charge is distributed between the two capacitors is

- (A) $\frac{1}{4} CV_0^2$
- (B) $\frac{1}{3} CV_0^2$
- (C) $\frac{1}{6} CV_0^2$
- (D) $\frac{1}{2} CV_0^2$

Q5 JEE Main 2020 - 5 September (Morning)

Two capacitors of capacitances C and $2C$ are charged to potential differences V and $2V$, respectively. These are then connected in parallel in such a manner that the positive terminal of one is connected to the negative terminal of the other. The final energy of this configuration is

- (A) $\frac{3}{2} CV^2$
- (B) $\frac{9}{2} CV^2$
- (C) Zero
- (D) $\frac{25}{6} CV^2$

Q6 JEE Main 2020 - 5 September (Evening)

A parallel plate capacitor has plate of length is connected to a battery of emf V . A dielectric slab of the same thickness 'd' and of dielectric constant $k = 4$ is being inserted between the plates of the capacitor. At what length of the slab inside plates, will the energy stored in the capacitor be two times the initial energy stored?

- (A) $\frac{\ell}{4}$
- (B) $\frac{\ell}{2}$
- (C) $\frac{2\ell}{3}$
- (D) $\frac{\ell}{3}$

Q7 JEE Main 2020 - 5 September (Evening)

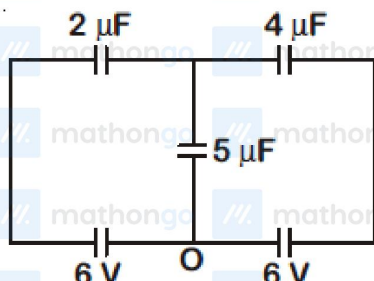
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Physics

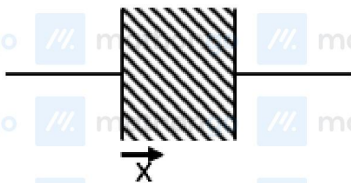
In the circuit shown, charge on the $5\mu\text{F}$ capacitor is



- (A) $16.36\mu\text{C}$
- (B) $18.00\mu\text{C}$
- (C) $5.45\mu\text{C}$
- (D) $10.90\mu\text{C}$

Q8 JEE Main 2020 - 7 January (Morning)

A parallel plate capacitor has plates of area A separated by distance d between them. It is filled with a dielectric which has a dielectric constant that varies as $k(x) = k_0(1 + \alpha x)$ where x is the distance measured from one of the plates. If $(\alpha d) < 1$, the total capacitance of the system is best given by the



expression:

- (A) $\frac{k_0\epsilon_0 A}{d} \left(1 + \alpha^2 d^2\right)$
- (B) $\frac{k_0\epsilon_0 A}{d} \left(1 + \frac{\alpha d}{2}\right)$
- (C) $\frac{k_0\epsilon_0 A}{2d} (1 + \alpha d)$
- (D) $\frac{k_0\epsilon_0 A}{2d} \left(1 + \frac{\alpha d}{2}\right)$

Q9 JEE Main 2020 - 7 January (Evening)

A 60pF capacitor is fully charged by a 20V supply. It is then disconnected from the supply and is connected to another uncharged 60pF capacitor in parallel. The electrostatic energy that is lost in this

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Physics

process by the time the charge is redistributed between them is (in nJ)

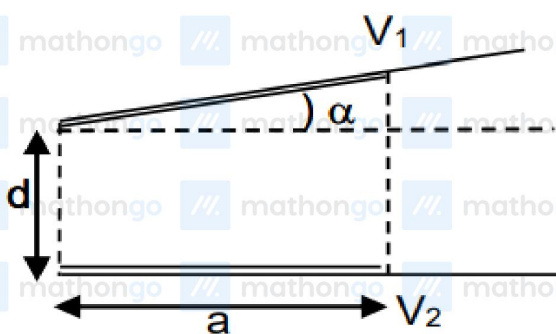
Q10 JEE Main 2020 - 8 January (Morning)

Effective capacitance of parallel combination of two capacitors C_1 and C_2 is $10\mu F$. When these capacitors are individually connected to a voltage source of $1V$, the energy stored in the capacitor C_2 is 4 times that of C_1 . If these capacitors are connected in series, their effective capacitance will be:

- (A) $1.6\mu F$
- (B) $16\mu F$
- (C) $4\mu F$
- (D) $\frac{1}{4}\mu F$

Q11 JEE Main 2020 - 8 January (Evening)

A capacitor is made of two square plates each of side a making a very small angle α between them, as shown in figure. The capacitance will be close to :



- (A) $\frac{\epsilon_0 a^2}{d} \left(1 - \frac{\alpha a}{2d}\right)$
- (B) $\frac{\epsilon_0 a^2}{d} \left(1 - \frac{\alpha a}{d}\right)$
- (C) $\frac{\epsilon_0 a^2}{d} \left(1 + \frac{\alpha a}{2d}\right)$
- (D) $\frac{\epsilon_0 a^2}{d} \left(1 - \frac{\alpha a}{4d}\right)$

Q12 JEE Main 2020 - 9 January (Evening)

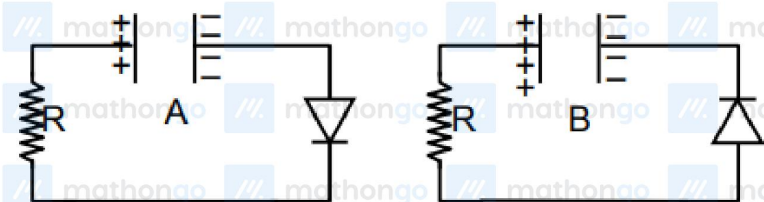
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Physics

Two identical capacitors A and B , charged to the same potential $5V$ are connected in two different circuits as shown below at time $t = 0$. If the charge on capacitors A and B at time $t = CR$ is Q_A and Q_B respectively, then (Here e is the base of natural logarithm)



- (A) $Q_A = VC$, $Q_B = CV$
- (B) $Q_A = VC$, $Q_B = VC/e$
- (C) $Q_A = VC/e$, $Q_B = CV/2$
- (D) $Q_A = CV/2$, $Q_B = VC/e$

Capacitance

Questions with Answer Keys

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Physics

Answer Key

Q1 (4)

Q2 (C)

Q3 (B)

Q4 (C)

Q5 (A)

Q6 (D)

Q7 (A)

Q8 (B)

Q9 (6)

Q10 (A)

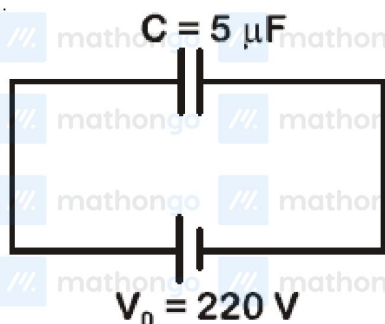
Q11 (A)

Q12 (B)

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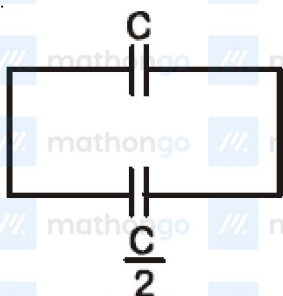
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Q1



$$Q_0 = CV_0$$

Now battery is disconnected and a capacitor of $\frac{C}{2}$ is connected to C then



Total charge = Q_0

$$C_{eq} = \frac{3C}{2}$$

$$\therefore \Delta E = \frac{Q_0^2}{2C} - \frac{2Q_0^2}{3C \times 2} \Rightarrow \Delta E = \frac{Q_0^2}{2C} \left(\frac{1}{3} \right)$$

$$\Rightarrow \Delta E = \frac{1}{6} \times 220 \times 220 \times 5 \times 10^{-6} \approx 4 \times 10^{-2} \text{ J}$$

$$\therefore X = 4$$

Q2

$$V_{\text{final}} = \frac{C_1 V}{C_1 + C_2}$$

$$C_2 = 15 \mu\text{F}$$

Capacitance

Hints & Solutions

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Physics

Q3

$$Q_3 = C_3 V_3$$
$$V_3 = 20\text{V}$$

$$Q_3 = 160\mu\text{C}$$

$$Q = Q_2 + Q_3$$

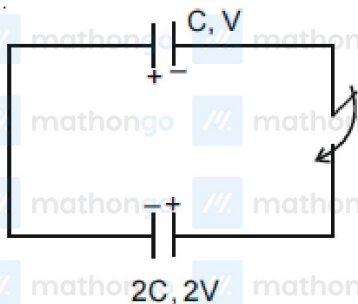
$$Q_2 = 590\mu\text{C}$$

Q4

$$\Delta U = \frac{1}{2} \frac{C_1 C_2}{C_1 + C_2} V_0^2 = \frac{1}{6} C V_0^2$$

Q5

$$V_{\text{common}} = \frac{4CV - CV}{3C} = V$$



$$C_{eq} = C + 2C = 3C$$

$$\therefore U_f = \frac{1}{2} \times (3C) \times V^2 = \frac{3}{2} CV^2$$

Q6

$$U = \frac{1}{2} CV^2$$

$$\frac{U_1}{U_2} = 2 \Rightarrow \frac{C_1}{C} = 2$$

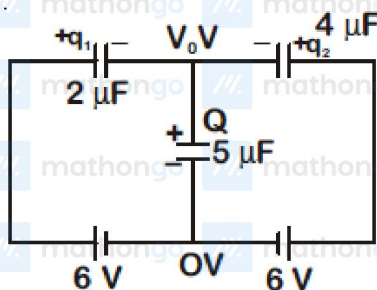
$$\frac{kx + (\ell - x)}{\ell} = 2$$

$$x = \frac{\ell}{3}$$

Q7

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$$Q = q_1 + q_2$$

$$5V_0 = 2(6 - V_0) + 4(6 - V_0)$$

$$V_0 = \frac{36}{11} \text{ V}$$

$$Q = 5V_0 = \frac{180}{11} \mu\text{C}$$

Q8

$$\text{Capacitance of element} = \frac{k\varepsilon_0 A}{dx}$$

$$\text{Capacitance of element } C' = \frac{k_0(1 + \alpha x)\varepsilon_0 A}{dx}$$

$$\sum \frac{1}{C'} = \int_0^d \frac{dx}{k_0\varepsilon_0 A(1 + \alpha x)}$$

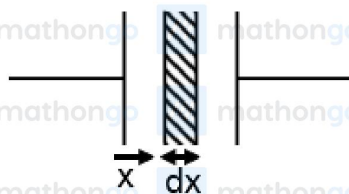
$$\frac{1}{C} = \frac{1}{k_0\varepsilon_0 A\alpha} \ln(1 + \alpha d)$$

Given $-\alpha d \ll 1$

$$\frac{1}{C} = \frac{1}{k_0\varepsilon_0 A\alpha} \left(\alpha d - \frac{\alpha^2 d^2}{2} \right)$$

$$\frac{1}{C} = \frac{d}{k_0\varepsilon_0 A} \left(1 - \frac{\alpha d}{2} \right)$$

$$C = \frac{k_0\varepsilon_0 A}{d} \left(1 + \frac{\alpha d}{2} \right)$$



Q9



$$V_0 = 20 \text{ V}$$

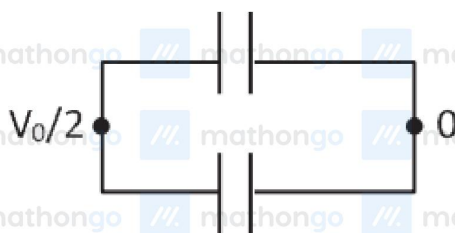
$$\text{Heat loss} = U_i - U_f$$

$$= \frac{1}{2} C V_0^2 - 2 \left[\frac{1}{2} C \left(\frac{V_0}{2} \right)^2 \right]$$

$$= \frac{C V_0^2}{4}$$

$$= \frac{(60 \times 10^{-12}) (20)^2}{4} \text{ J}$$

$$= 6 \times 10^{-9} \text{ J} = 6 \text{ nJ}$$



Q10

$$\text{Given } C_1 + C_2 = 10 \mu\text{F} \dots (i)$$

$$4 \left(\frac{1}{2} C_1 V^2 \right) = \frac{1}{2} C_2 V^2$$

$$\Rightarrow 4C_1 = C_2 \dots (ii)$$

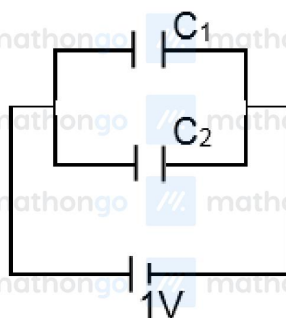
from equation (i) & (ii)

$$C_1 = 2 \mu\text{F}$$

$$C_2 = 8 \mu\text{F}$$

If they are in series

$$C_{eq} = \frac{C_1 C_2}{C_1 + C_2} = 1.6 \mu\text{F}$$

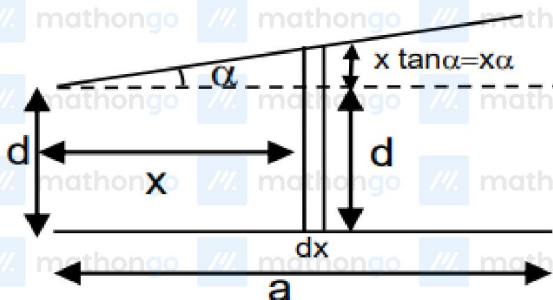


Q11

$$dc = \frac{\epsilon_0 a dx}{d + \alpha x}$$

$$\Rightarrow c = \frac{\epsilon_0 a}{\alpha} [\ln(d + \alpha x)]_0^a$$

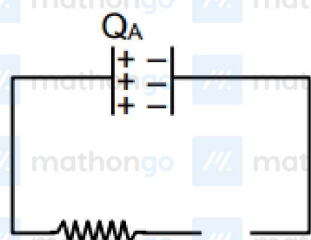
$$= \frac{\epsilon_0 a}{\alpha} \ln\left(1 + \frac{\alpha a}{d}\right) \approx \frac{\epsilon_0 a^2}{d} \left(1 - \frac{\alpha a}{2d}\right)$$



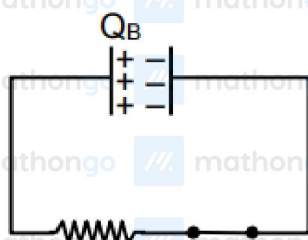
Q12

Maximum charge on capacitor = $5CV$

(a) is reverse biased and (b) is forward biased
for case (a)



$$Q_A = CV$$



$$\text{so } q = q_{\max} [e^{-t/RC}]$$

$$Q_B = CVe^{-1}$$